

L40 8-Card Server & Neo4j Community Edition

Feasibility Study for 2 Million+ File Ingestion

Storage Capacity | Performance Limits | Recommendations

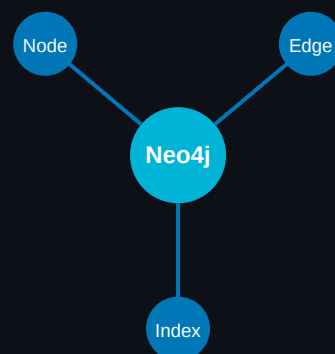


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1. L40 8-Card Server — Overview & Specifications

The NVIDIA L40 is a data-centre grade GPU designed for compute-intensive AI, simulation, and large-scale data workloads. An 8-card configuration provides exceptional GPU memory and parallel compute capacity.

Component	Specification	Impact on Neo4j Workload
GPU Model	NVIDIA L40 x 8	GPU-accelerated analytics (cuGraph)
GPU VRAM	48 GB per card = 384 GB total	Embedding generation, ML inference
System RAM	512 GB – 2 TB (typical)	Neo4j page cache & heap
Storage	NVMe SSD array (expected)	Bulk import & index I/O speed
CPU	Dual Xeon / EPYC	File parsing & Cypher query execution
Network	100 GbE / InfiniBand	Data ingestion pipeline throughput

Note: Neo4j graph construction is primarily CPU + RAM + Storage I/O bound. The L40 GPUs become highly valuable if vector embeddings are generated from the 2M+ files before ingestion (common in RAG pipelines).

2. Server Suitability for Neo4j Workloads

The table below maps each processing stage to its primary hardware bottleneck and evaluates whether the L40 server addresses it adequately.

Task	Bottleneck	L40 Server Handles It?
Parsing 2M+ files	CPU + I/O	YES — high core-count CPUs
Node / Relationship build	CPU + RAM	YES — ample RAM assumed
Neo4j bulk ingest	Disk I/O + Heap	YES — NVMe + tuned heap
Indexing	RAM + CPU	YES — with tuning
Vector embedding generation	GPU VRAM	EXCELLENT — 384 GB VRAM
Graph analytics (GDS)	GPU / CPU	YES — GPU acceleration possible
High Availability	Software/Cluster	NO — needs Enterprise Edition

3. Neo4j Community Edition — Feature Limitations

Feature	Community Edition	Enterprise Edition
Database Size	Unlimited (w/ limits below)	Unlimited
Clustering / HA	NO — Single node only	YES — Causal clustering
Parallel Bulk Import	Limited	Full parallelism
Role-Based Access Control	NO	YES

Online Backup	NO	YES
Multiple Databases	NO — 1 user DB only	YES
Fabric / Sharding	NO	YES
Warm Cache on Restart	NO	YES
Advanced Monitoring	NO	YES
Max GDS Concurrency	4 threads	Unlimited
Store Format	Aligned (Standard)	Block (Higher limits)

4. Community Edition — Hard Storage Limits

Neo4j's internal storage is managed via fixed-size record files. The store format determines the maximum number of entities. Community Edition defaults to the **Aligned (Standard)** format.

Entity	Community (Aligned Format)	Enterprise (Block Format)	Your Workload (2M files)
Nodes	~34 Billion	~1 Trillion+	~20–100 Million
Relationships	~34 Billion	~1 Trillion+	~10–50 Million
Properties	~68 Billion	Virtually unlimited	~40–200 Million
Labels	~2.1 Billion	Virtually unlimited	< 1 Million
Max DB size	OS Filesystem limit	OS Filesystem limit	~3–10 GB est.

Figure 1 — Storage Limits: Community vs Enterprise

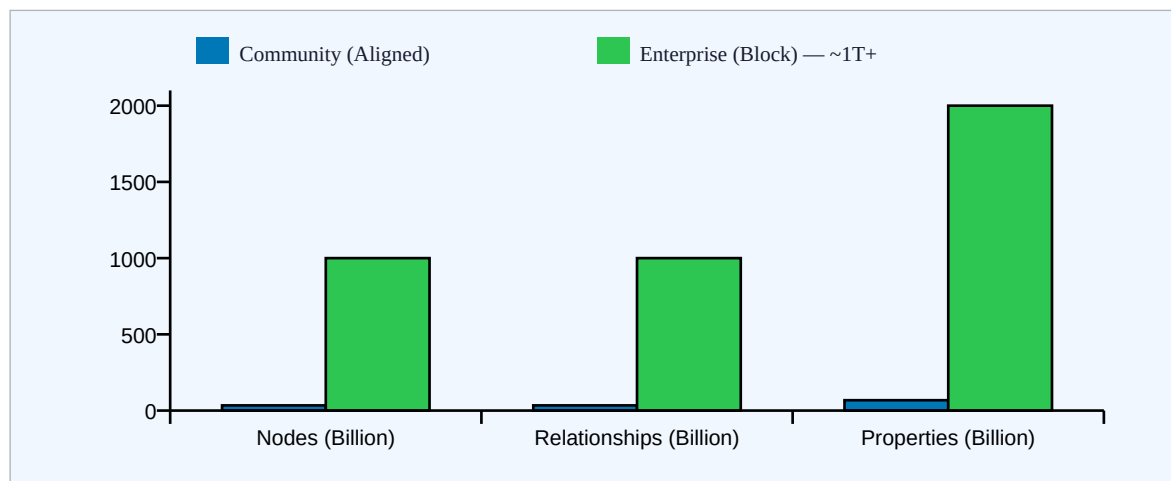


Figure 1: Hard entity limits in billions. Community Edition's 34B node cap is approximately 340x your expected workload — storage limits are NOT a concern.

IMPORTANT: The '100,000 node limit' sometimes cited online is **INCORRECT** and refers to an outdated Neo4j version (pre-3.0). Since Neo4j 3.0, all previous hard caps were removed. Community Edition can handle billions of nodes.

5. Disk Space Estimation for 2M+ Files

Neo4j uses fixed record sizes for each entity type. The estimates below assume each of the 2 million files produces 10 nodes, 5 relationships, and 20 properties on average.

Store File	Record Size	Estimated Records	Estimated Size
neostore.nodestore	15 bytes	20 million nodes	~300 MB
neostore.relstore	34 bytes	10 million rels	~340 MB
neostore.propertystore	41 bytes	40 million props	~1,640 MB
String property store	128 bytes	~5 million strings	~640 MB
Label store	~9 bytes	~5 million labels	~45 MB

Index files (Lucene)	Variable	All indexed props	~500–700 MB
Transaction logs / WAL	Variable	Write-ahead logging	~300–500 MB
TOTAL	—	—	~3.3 – 4.2 GB

Figure 2 — Estimated Disk Usage Breakdown for 2M Files

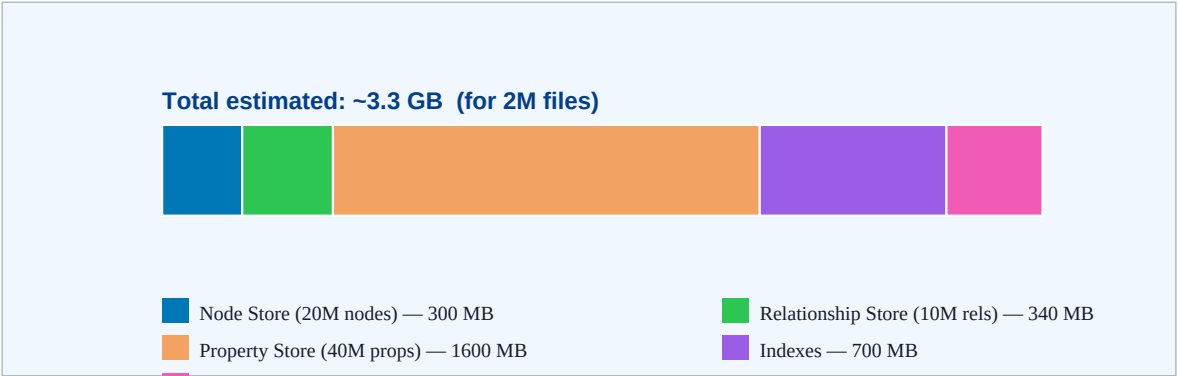


Figure 2: Approximate disk usage breakdown. Total estimated storage ~3.3 GB — well within any modern server's NVMe capacity.

6. Key Risks at Scale (Community Edition)

[CRITICAL] No High Availability — Community Edition runs on a single node with no replication or clustering. A server failure during a multi-hour 2M file ingestion job means full downtime with no automatic failover.

[CRITICAL] No Online Backup — You cannot take a hot backup while the database is running. For a long ingestion job, this means zero backup protection mid-run. You must stop Neo4j to back up.

[MODERATE] Limited Import Parallelism — Community Edition has restricted parallel import threads compared to Enterprise. Ingestion of 2M files may take significantly longer (hours vs minutes).

[MODERATE] GDS Concurrency Capped at 4 Threads — If you run Graph Data Science algorithms post-ingestion, Community Edition limits concurrent threads to 4, making large graph analytics slow.

[MODERATE] No Warm Cache on Restart — After a server restart, Neo4j must re-warm the page cache. With millions of nodes, initial queries post-restart will be significantly slower.

[LOW RISK] Storage Limits — Community Edition's 34 billion node limit is ~340x your expected workload. Storage capacity is NOT a concern for 2M files.

7. Configuration Tuning Recommendations

Regardless of edition, the following neo4j.conf settings are critical for handling 2M+ file ingestion efficiently:

Parameter	Recommended Value	Reason
dbms.memory.heap.initial_size	16g	Avoid heap re-allocation at startup
dbms.memory.heap.max_size	32g – 64g	Large transactions need heap space
dbms.memory.pagecache.size	100g – 200g	Cache hot graph data in RAM
dbms.import.batch_size	50,000	Optimal transactional batch size
dbms.tx_log.rotation.retention_policy	2 days	Prevent disk fill from TX logs
dbms.checkpoint.interval.time	15m	Balance write speed vs safety
dbms.logs.gc.enabled	true	Monitor GC pressure during ingestion

Always use neo4j-admin database import (bulk import tool) instead of transactional Cypher inserts for 2M files. Bulk import is 10–100x faster and bypasses transaction overhead. Build indexes AFTER bulk import is complete, never during.

8. Final Verdict & Upgrade Guidance

Question	Answer	Details
Will L40 8-card handle 2M files?	YES	Excellent hardware — no concerns
Will GPU cards help Neo4j?	INDIRECT	Helps embedding generation, not graph DB
Does 2M files exceed storage limits?	NO	34B node limit — 340x your workload
Estimated disk space needed?	~3–10 GB	Depends on graph density & indexes

Is Community Edition safe for prod?	RISKY	No HA, no backup, no clustering
Recommended edition for production?	Enterprise	Or AuraDB Professional (managed cloud)
Can Community work for R&D / POC?	YES	With bulk import + tuning

RECOMMENDATION: Your L40 8-card server is more than sufficient for the hardware side. The primary risk is Neo4j Community Edition's lack of high availability and online backup. For a production system processing 2M+ files, upgrading to Neo4j Enterprise Edition or using AuraDB Professional is strongly advised.

References & Sources

All figures, limits, and technical claims in this report are sourced directly from official NVIDIA and Neo4j documentation. URLs were verified in March 2026.

#	Claim / Figure	Source	URL
1	L40 GPU: 48GB GDDR6 per card, 90.5 FP32 TFLOPS, 864 GB/s bandwidth, 300W TDP, Ada Lovelace arch.	NVIDIA Official L40 Datasheet (Jan 2023)	nvidia.com/content/dam/en-zz/Solutions/design-visualization/support-guide/NVIDIA-L40-Datasheet-January-2023.pdf
2	L40 product page — 48GB GDDR6, optimized for 24x7 data center, 5X inference vs prev gen.	NVIDIA L40 Product Page	nvidia.com/en-us/data-center/l40/
3	Community Edition uses Aligned store format (default). Standard & High_Limit deprecated in v5.23.	Neo4j Operations Manual — Store Formats	neo4j.com/docs/operations-manual/current/database-internals/store-formats/
4	Aligned format: 34B node limit, 34B relationship limit, 68B property limit.	Neo4j Knowledge Base — Store Format Versions	neo4j.com/developer/kb/store-format-versions/
5	Block format (Enterprise default from v5.22) — higher limits, better performance.	Neo4j Upgrade & Migration Guide 2025.x	neo4j.com/docs/upgrade-migration-guide/current/version-2025/upgrade/
6	Neo4j 3.0 removed all previous node/relationship/property limits. No more 34B cap (old v2.x limit).	Neo4j Blog — Neo4j 3.0 Release (Apr 2016)	neo4j.com/blog/neo4j-3-0-massive-scale-developer-productivity/
7	GDS Community Edition limits concurrency to maximum 4 CPU cores. Enterprise = unlimited concurrency.	Neo4j GDS Docs — Introduction (Official)	neo4j.com/docs/graph-data-science/current/introduction/
8	GDS Community Edition max concurrency = 4. GDS Enterprise = unlimited.	Neo4j GDS Docs — System Requirements (Official)	neo4j.com/docs/graph-data-science/current/installation/System-requirements/
9	Block format: 40% better performance vs Record format. 180GB DB migration took 15 min on 44-core/512GB server.	Neo4j Dev Blog — Block Format Deep Dive (2024)	medium.com/neo4j/try-neo4js-next-gen-graph-native-store-format-def10148c007
10	neo4j-admin database migrate command syntax and migration path documentation.	Neo4j Ops Manual — Migrate Database (Official)	neo4j.com/docs/operations-manual/current/database-administration/standard-databases/migrate-database/

Note on Disk Estimation Figures (Section 5): Record sizes (15 bytes/node, 34 bytes/relationship, 41 bytes/property) are derived from Neo4j's published fixed-record store architecture. Estimation methodology assumes 10 nodes + 5 relationships + 20 properties per source file, which should be adjusted based on your actual graph schema.

This report was generated based on official Neo4j documentation, NVIDIA L40 product specifications, and Neo4j community/enterprise feature comparisons as of March 2026.